

Fourier analysis – 2

Fourier coefficients and smoothness

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Theoretical minimum.

Theorem 1 (differentiation and decay). For an absolutely continuous f one has $c_n(f') = in c_n(f)$; hence $f \in C^k$ forces $c_n = o(n^{-k})$, and conversely $\sum |n|^k |c_n| < \infty$ forces $f \in C^k$. Through Parseval the exact statement is $\sum |n|^{2k} |c_n|^2 = \frac{1}{2\pi} \int |f^{(k)}|^2$: smoothness is the decay of coefficients measured in the mean square.

Theorem 2 (Hölder and the shift trick). If $|f(x) - f(y)| \leq C|x - y|^\alpha$, then $|c_n| \leq \frac{C}{2} \left(\frac{\pi}{|n|}\right)^\alpha$. The proof is the substitution $x \mapsto x + \frac{\pi}{n}$, which changes the sign of e^{inx} and turns the coefficient into an integral of a difference — the standard device for every estimate of this kind.

Theorem 3 (Bernstein). If f is Hölder of order $\alpha > \frac{1}{2}$, then $\sum |c_n| < \infty$, so the Fourier series converges absolutely and uniformly. The exponent $\frac{1}{2}$ is sharp, and the proof is Cauchy–Bunyakovsky applied dyadically to the estimate of Theorem 2 — a model of how a rate of approximation is converted into summability.

Fact 1 (what the coefficients cannot do). Coefficients of an L^1 function tend to zero (Riemann–Lebesgue) but may do so arbitrarily slowly; not every null sequence is a sequence of Fourier coefficients, and for odd coefficients the necessary condition $\sum \frac{b_n}{n} < \infty$ rules out, for example, $\sum \frac{\sin nx}{\ln n}$.

Fact 2 (structure from coefficients). Vanishing of all negative coefficients, convexity or positivity of the coefficient sequence, and lacunarity of its support are three hypotheses that force strong structural conclusions about the function; the extreme case is a finite spectrum, which means that f is a trigonometric polynomial.

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1. Prove $c_n(f') = inc_n(f)$ and both directions of Theorem 1, including the Parseval form; then show that $c_n = O(n^{-k})$ alone does not imply $f \in C^{k-1}$.
 2. (Kаpacёв 1.71) Prove that a function satisfying $|f(x) - f(y)| \leq C|x - y|^\alpha$ on $[-\pi, \pi]$ has Fourier coefficients with $|c_n(f)| \leq C\left(\frac{\pi}{n}\right)^\alpha$, and prove Bernstein's theorem for $\alpha > \frac{1}{2}$.
 3. (Kаpacёв 1.72) Find the exact order of decay of the Fourier coefficients of $f(x) = \sqrt{|x|}$ on $[-\pi, \pi]$.
 4. (Kаpacёв 1.70) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be 2π -periodic and infinitely differentiable and put $m_n = \left(\int_0^{2\pi} |f^{(n)}(x)|^2 dx\right)^{1/2}$. Prove that the sequence $(m_n^{1/n})$ has a limit, finite or infinite, and describe the functions for which the limit is finite.
 5. (Kаpacёв 1.73) Suppose that for a continuous 2π -periodic f the set of its translates $\{f(x+t)\}_{t \in \mathbb{R}}$ spans a finite-dimensional subspace of $C(\mathbb{R})$. Prove that f is a trigonometric polynomial.